UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2007 question paper

5054 PHYSICS

5054/04 Paper 4 (Alternative to Practical), maximum raw mark 30

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International Examinations

1 **(a)** (allow) to reach terminal/constant velocity/stop accelerating/forces balance/ speed/velocity not constant at start/near surface/speed zero at surface B1
allow only constant velocity between bands
not velocity decreases

(b) reduces parallax error/see when sphere passes (behind) band/ bands show when to start and stop stopwatch/distance measured accurately B1
allow bands act as a reference point/mark(er)/fiducial mark(er)
ball touches/hits band
time/distance measured easily

(c) yes/almost B1
comments e.g. doubling/halving x doubles/halves t/x directly proportional to t
$x = kt$ or $^x/_t = k$ or $^t/_x = k$
or values quoted
e.g. velocities calculated both 4.6 or 4.63 and 4.65 or approximately 5
ratio found 40/20 = 2 and 8.64/4.60 = 2.(0093)
check table doubling t gives 8.60, nearly same as 8.64
halving t gives 4.32, nearly same as 4.30
finding k: 40/8.64 = 4.63, 20/4.30 = 4.65
or 8.64/40 = 0.216, 4.30/20 = 0.215 B1

allow for one mark no + comment on table values
no + ratios found and not the same
no + difference is 0.04 s
not just velocities are the same without evidence

(d) **(i)** eye level horizontal and level with top band
accept E/dot/ at side or on top of jar/eye drawn looking towards jar B1

(ii) vertical ruler drawn on jar/no further than x marker arrow from jar
accept rectangle with no labels or marks/line with marks or label B1

(e) **(i)** all values to one d.p.(even if incorrect) B1
any 2 answers in the range 10.5 → 10.6; 18.1 → 18.2; 41.6 → 41.7 (3 or more sf) C1
10.6; 18.2; 41.7 all correct allow 3 or more sf (10.582; 18.182; 41.667) A1
allow inconsistent sf

(ii) for one mark: doubling/halving d does not double/halve v
statement of requirement for directly proportional
d not = kv, $^d/_v$ or $^v/_d$ not constant
graph of d against v not a straight line through 0,0 C1

for two marks: doubling d gives 4 x v
clear use of data from table to support statement A1
allow sketch graph
ecf table increasing d/v decreases v/d
not comparing d or v with t/inversely proportional/v x d not constant
explanation of linearity (e.g. rate of increase not same)

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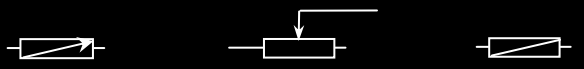
(iii) diameter of sphere; metal used for sphere; type of oil and none wrong ✓ ✓ ✓ B2
any two correct and only one wrong scores one mark

[Total: 13]

2 (a) ammeter and d.c. power supply in series B1
rheostat/variable resistor and d.c. power supply in series/
variable power supply B1
ignore additional components if circuit works
e.g. additional resistors/voltmeter correctly connected
allow switch open or closed
for power supply: —o o—
for resistor: thermistor/LDR



labelled rheostat

not

max 1 mark if any components drawn over circuit wire/additional incorrectly connected components

allow for one mark: correct ammeter and variable resistor with incorrect power supply
no marks: no power supply shown

(b) (i) magnetic field small/increases field/more field lines/to attract the iron filings
for the force to be reasonable/so the magnetic field can be detected B1
not more fields/so iron filings are magnetised

(ii) help iron filings align with the field/reduce friction
overcome inertia/filings large or heavy B1
allow to show the field/so iron filings spread out/
so iron filings can move

(iii) easier to move/easier to attract/show field more accurately/lighter weight/
less mass or inertia/more sensitive to the field/clearer field/
follow field without overlapping B1
allow show weaker field
ignore more easily/quickly magnetised

[Total: 5]

3 (a) (i) incident ray continued straight with ruler (labelled XY) B1
allow dotted line
line to level with P_4

(ii) (refracted and) emergent rays correctly drawn with ruler B1

(iii) *(d)* marked correctly between lines B1
allow line not accurate but intention clear
not horizontal line
if d varies significantly

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(iv) *d* = 0.7 to 1.2 cm to nearest mm unit required
ecf **(a) (iii)** for correct attempt at d B1
allow 0.05 cm
not to 0.01 cm e.g. 0.90 cm, 0.92 cm

(b) 5.0 cm/50 mm (correct answer only) unit required unless penalised in **(a) (iv)** B1
allow 5 cm no s.f. penalty

(c) with t = 5 cm

d	n
0.7	1.39
0.8	1.47
0.85	1.52
0.9	1.56
0.95	1.61
1.0	1.67
1.1	1.79
1.2	1.92

calculation correct to 2 d.p. ignore subsequent rounding B1

allow ecf **(a) (iii)** and **(b)**
not negative answer
unit given (e.g. °)

[Total: 6]

4 **(a)** (when d = 0) band is not stretched/no force on block/block not pulled/moved yet/
band has no energy B1

(b) table for graph

5	13.3
6	20.9
7	28.7
8	39.6
9	51.8
10	62.5

axes: – correct way round, labelled quantity and unit ↑ D/cm → d/cm B1
scales: – sensible, more than ½ grid/from (0,0) B1
points: plotted accurately (within ½ small square) and neat B1
not if scale not sensible
line: best fit smooth curve, neat B1

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(c) any sensible practical statement B1

e.g. (tape) ruler on bench/block slides along ruler/repeat experiment/
avoid parallax when reading (ruler)/used a thread/tape measure/string/
mark block position on floor/calibrate bench/

allow annotation on diagram page 8

not use a stopwatch/time block

[Total: 6]

Marking Scheme Code

B1 Independent mark

C1 Compensation mark:
– is always followed by an A mark
– is given automatically if the answer is correct
– is given if the answer is wrong but the point is seen in the working

A1 Answer mark

ecf error carried forward; correct working using an error in previous working is credited

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